CASE STUDY 6

SwiftMotors & IBM Cognos Analytics

Automobile Manufacturing

Q2 Downtime Analysis Report

Build a Downtime Analysis Report listing Assembly Lines with total downtime minutes per month.

- **Requirement:** Add a prompt for Plant and Shift selection.
- Sort by Total Downtime descending.
- Apply Conditional Styling: lines with downtime > 500 minutes highlighted in Red.
- Use Tahoma font, size 11. Summarize the total production hours lost due to downtime at the report footer.

Plant: Chennai **Shift:** Morning Shift

Downtime Analysis Report

Assembly Line	Total Downtime (min)
Line 4	850
Line 2	640
Line 5	640
Line 1	578
Line 3	652

Total Production Hours Lost: 45.2 hours

CASE STUDY QUESTION

Q2 [Reporting] Downtime Analysis Report

Build a **Downtime Analysis Report** listing Assembly Lines with total downtime minutes per month. Add a prompt for Plant and Shift selection. Sort by Total Downtime descending.

- Apply Conditional Styling: lines with downtime > 500 minutes highlighted in Red.
- Use Tahoma font, size 11. Summarize the total production hours lost due to downtime at the report footer.

Steps to Create the Report:

01 Set Up Report Layout

Design the report in IBM Cognos Analytics. Set up prompts to select the **Plant and Shift.** Use Tahoma font, size 11.

02 Build Downtime Analysis Report

List Assembly Lines with total downtime minutes per month. Sort by Total Downtime descending.

Plant

Shift

Downtime Analysis Report

Assembly Line Minutes		Total Downtime
Pune	750	**335 K**
Chennai	600	**297 K**
Ahmedabad	400	**278 K**

Total Downtime minutes: **3,324** minutes

03 Apply Conditional Styling

Highlight lines with downtime > 500 minutes in Red.

04 Summarize Total Downtime

Summarize the total production hours lost because of downtime at the report footer.

My IBM | My CS 2 | My CS 2

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&id=i428A00C9675248E8B5C2A0222938F188&objRef=i428A00C9675248E8B5C2A0222938F188...

IBM Cognos Analytics | My CS 2

Edit | Report > Pages > Page1 | Page design | Properties

Insertable objects

Find

PINNED

Text item, Block, Table, List, Crosstab, Visualization

TEXTUAL

Text item, Query calculation, Layout calculation, Rich text item, Hyperlink, Date, Time, Page number, Row number, As-of-time expression

LAYOUT

DATA CONTAINER

Data table, List, Crosstab, Visualization, Repeater table, Repeater, Singleton

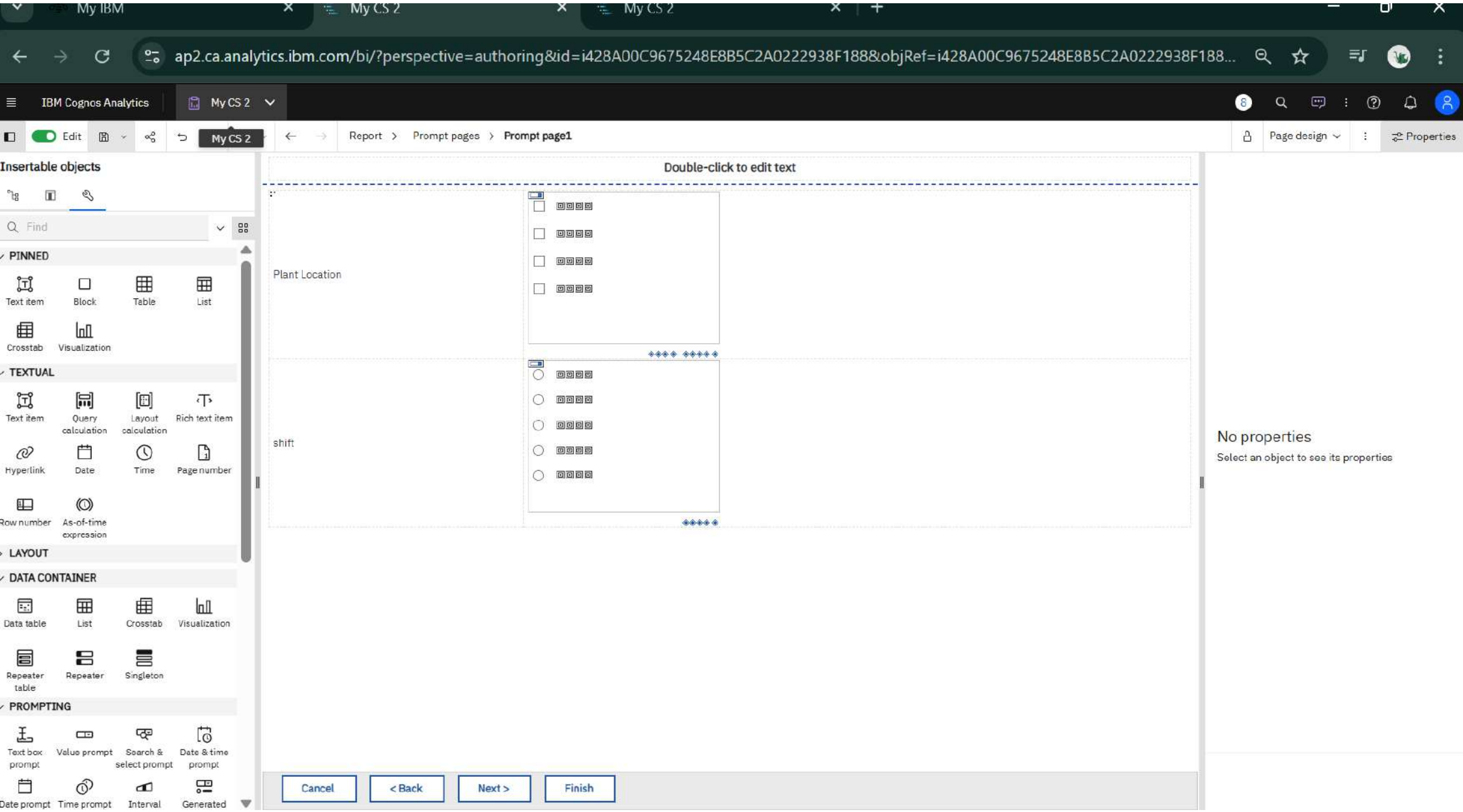
PROMPTING

Text box prompt, Value prompt, Search & select prompt, Date & time prompt, Date prompt, Time prompt, Interval, Generated

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Total Downtime Minutes	Total Production Hours Lost
<Plant Location>	<Shift>	<Assembly Line>	<Production Date>	<Report Month>	<Downtime (mins)>	<Total Downtime Minutes>	<Total Production Hours Lost>
<Assembly Line> - Total					<Total(Downtime (mins)1)>		
<Assembly Line> - Total					<Total(Downtime (mins))1>		
<Plant Location>	<Shift>	<Assembly Line>	<Production Date>		<Downtime (mins)>	<Total Downtime Minutes>	<Total Production Hours Lost>
<Assembly Line> - Total					<Total(Downtime (mins)1)>		
<Assembly Line> - Total					<Total(Downtime (mins))1>		
<Report Month> - Total					<Total(Downtime (mins)1)>		
<Report Month> - Total					<Total(Downtime (mins))1>		
<Plant Location>	<Shift>	<Assembly Line>	<Production Date>	<Report Month>	<Downtime (mins)>	<Total Downtime Minutes>	<Total Production Hours Lost>
<Assembly Line> - Total					<Total(Downtime (mins)1)>		
<Assembly Line> - Total					<Total(Downtime (mins))1>		
<Plant Location>	<Shift>	<Assembly Line>	<Production Date>		<Downtime (mins)>	<Total Downtime Minutes>	<Total Production Hours Lost>
<Assembly Line> - Total					<Total(Downtime (mins)1)>		
<Assembly Line> - Total					<Total(Downtime (mins))1>		
<Report Month> - Total					<Total(Downtime (mins)1)>		
<Report Month> - Total					<Total(Downtime (mins))1>		
<Total Downtime Minutes> - Total					<Total(Downtime (mins)1)>		
<Total Downtime Minutes> - Total					<Total(Downtime (mins))1>		

No properties

Select an object to see its properties


My IBM
My CS 2
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IBM Cognos Analytics
My CS 2
Edit
My CS 2
Report > Prompt pages > Prompt page1
Page design
Properties
Insertable objects
Find
PINNED
Text item
Block
Table
List
Crosstab
Visualization
TEXTUAL
Text item
Query calculation
Layout calculation
Rich text item
Hyperlink
Date
Time
Page number
Row number
As-of-time expression
LAYOUT
DATA CONTAINER
Data table
List
Crosstab
Visualization
Repeater table
Repeater
Singleton
PROMPTING
Text box prompt
Value prompt
Search & select prompt
Date & time prompt
Date prompt
Time prompt
Interval
Generated
Double-click to edit text
Plant Location
shift
Cancel
< Back
Next >
Finish
No properties
Select an object to see its properties

My IBM | My CS 2 | My CS 2

ap2.ca.analytics.ibm.com/bi/?perspective=authoring&closeWindowOnLastView=true

Maintenance: The upgrade is now complete. Click on More Info to see what actions may be necessary and to subscribe to future events

Dismiss More info

IBM Cognos Analytics My CS 2

HTML

Mode: Interactive View Filters

Plant Location

- [x] Ahmedabad
- [ ] Chennai
- [x] Pune

Select all Deselect all

shift

- (•) Afternoon
- ( ) Morning
- ( ) Night

Deselect all

Cancel < Back Next > Finish

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
Pune	Afternoon	LINE-4	Sep 3, 2024	9	175	175	2.91666667
LINE-4 - Total					**175**		
LINE-4 - Total					**175**		
9 - Total					**175**		
9 - Total					**175**		
175 - Total					**175**		
175 - Total					**175**		
Pune	Afternoon	LINE-6	Mar 9, 2024	3	173	173	2.88333333
LINE-6 - Total					**173**		
LINE-6 - Total					**173**		
3 - Total					**173**		
3 - Total					**173**		
173 - Total					**173**		
173 - Total					**173**		
Pune	Afternoon	LINE-6	Dec 8, 2024	12	172	172	2.86666667
LINE-6 - Total					**172**		
LINE-6 - Total					**172**		
12 - Total					**172**		
12 - Total					**172**		
172 - Total					**172**		
172 - Total					**172**		
Ahmedabad	Afternoon	LINE-2	Jul 19, 2024	7	170	170	2.83333333
LINE-2 - Total					**170**		
LINE-2 - Total					**170**		
7 - Total					**170**		
7 - Total					**170**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
170 - Total					**170**		
170 - Total					**170**		
Pune	Afternoon	LINE-6	Apr 30, 2024	4	169	169	2.81666667
LINE-6 - Total					**169**		
LINE-6 - Total					**169**		
4 - Total					**169**		
4 - Total					**169**		
169 - Total					**169**		
169 - Total					**169**		
Pune	Afternoon	LINE-6	Nov 18, 2024	11	168	168	2.8
LINE-6 - Total					**168**		
LINE-6 - Total					**168**		
11 - Total					**168**		
11 - Total					**168**		
168 - Total					**168**		
168 - Total					**168**		
Pune	Afternoon	LINE-3	Jan 19, 2024	1	164	164	2.73333333
LINE-3 - Total					**164**		
LINE-3 - Total					**164**		
1 - Total					**164**		
1 - Total					**164**		
Pune	Afternoon	LINE-8	Dec 18, 2024	12	164	164	2.73333333
LINE-8 - Total					**164**		
LINE-8 - Total					**164**		
12 - Total					**164**		
12 - Total					**164**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
164 - Total					**328**		
164 - Total					**328**		
Ahmedabad	Afternoon	LINE-3	Jun 13, 2024	6	162	162	2.7
LINE-3 - Total					**162**		
LINE-3 - Total					**162**		
6 - Total					**162**		
6 - Total					**162**		
162 - Total					**162**		
162 - Total					**162**		
Pune	Afternoon	LINE-6	Aug 17, 2024	8	161	161	2.68333333
LINE-6 - Total					**161**		
LINE-6 - Total					**161**		
8 - Total					**161**		
8 - Total					**161**		
161 - Total					**161**		
161 - Total					**161**		
Pune	Afternoon	LINE-5	Apr 14, 2024	4	158	158	2.63333333
LINE-5 - Total					**158**		
LINE-5 - Total					**158**		
4 - Total					**158**		
4 - Total					**158**		
158 - Total					**158**		
158 - Total					**158**		
Pune	Afternoon	LINE-1	Feb 3, 2024	2	156	156	2.6
LINE-1 - Total					**156**		
LINE-1 - Total					**156**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
2 - Total					**156**		
2 - Total					**156**		
156 - Total					**156**		
156 - Total					**156**		
Pune	Afternoon	LINE-2	Apr 27, 2024	4	149	149	2.48333333
LINE-2 - Total					**149**		
LINE-2 - Total					**149**		
4 - Total					**149**		
4 - Total					**149**		
149 - Total					**149**		
149 - Total					**149**		
Pune	Afternoon	LINE-3	May 21, 2024	5	142	142	2.36666667
LINE-3 - Total					**142**		
LINE-3 - Total					**142**		
5 - Total					**142**		
5 - Total					**142**		
Ahmedabad	Afternoon	LINE-3	Jun 17, 2024	6	142	142	2.36666667
LINE-3 - Total					**142**		
LINE-3 - Total					**142**		
6 - Total					**142**		
6 - Total					**142**		
142 - Total					**284**		
142 - Total					**284**		
Ahmedabad	Afternoon	LINE-8	Feb 25, 2024	2	140	140	2.33333333
LINE-8 - Total					**140**		
LINE-8 - Total					**140**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
2 - Total					**140**		
2 - Total					**140**		
140 - Total					**140**		
140 - Total					**140**		
Ahmedabad	Afternoon	LINE-4	Apr 6, 2024	4	133	133	2.21666667
LINE-4 - Total					**133**		
LINE-4 - Total					**133**		
4 - Total					**133**		
4 - Total					**133**		
133 - Total					**133**		
133 - Total					**133**		
Ahmedabad	Afternoon	LINE-6	Jul 23, 2024	7	129	129	2.15
LINE-6 - Total					**129**		
LINE-6 - Total					**129**		
7 - Total					**129**		
7 - Total					**129**		
129 - Total					**129**		
129 - Total					**129**		
Ahmedabad	Afternoon	LINE-5	Mar 3, 2024	3	128	128	2.13333333
LINE-5 - Total					**128**		
LINE-5 - Total					**128**		
3 - Total					**128**		
3 - Total					**128**		
128 - Total					**128**		
128 - Total					**128**		
Ahmedabad	Afternoon	LINE-2	Jun 2, 2024	6	127	127	2.11666667

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
LINE-2 - Total				6	**127**		
LINE-2 - Total					**127**		
6 - Total					**127**		
6 - Total					**127**		
127 - Total					**127**		
127 - Total					**127**		
Pune	Afternoon	LINE-4	Jun 29, 2024	6	122	122	2.03333333
LINE-4 - Total					**122**		
LINE-4 - Total					**122**		
6 - Total					**122**		
6 - Total					**122**		
122 - Total					**122**		
122 - Total					**122**		
Pune	Afternoon	LINE-4	Nov 23, 2024	11	109	109	1.81666667
LINE-4 - Total					**109**		
LINE-4 - Total					**109**		
11 - Total					**109**		
11 - Total					**109**		
109 - Total					**109**		
109 - Total					**109**		
Ahmedabad	Afternoon	LINE-5	Feb 21, 2024	2	108	108	1.8
LINE-5 - Total					**108**		
LINE-5 - Total					**108**		
2 - Total					**108**		
2 - Total					**108**		
108 - Total					**108**		
108 - Total					**108**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
Ahmedabad	Afternoon	LINE-4	Sep 27, 2024	9	107	107	1.78333333
LINE-4 - Total					**107**		
LINE-4 - Total					**107**		
9 - Total					**107**		
9 - Total					**107**		
107 - Total					**107**		
107 - Total					**107**		
Pune	Afternoon	LINE-4	Apr 20, 2024	4	106	106	1.76666667
LINE-4 - Total					**106**		
LINE-4 - Total					**106**		
4 - Total					**106**		
4 - Total					**106**		
106 - Total					**106**		
106 - Total					**106**		
Ahmedabad	Afternoon	LINE-1	Feb 10, 2024	2	102	102	1.7
LINE-1 - Total					**102**		
LINE-1 - Total					**102**		
2 - Total					**102**		
2 - Total					**102**		
102 - Total					**102**		
102 - Total					**102**		
Pune	Afternoon	LINE-7	Feb 8, 2024	2	101	101	1.68333333
LINE-7 - Total					**101**		
LINE-7 - Total					**101**		
2 - Total					**101**		
2 - Total					**101**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
101 - Total					**101**		
101 - Total					**101**		
Ahmedabad	Afternoon	LINE-8	Jan 4, 2024	1	95	95	1.58333333
LINE-8 - Total					**95**		
LINE-8 - Total					**95**		
1 - Total					**95**		
1 - Total					**95**		
95 - Total					**95**		
95 - Total					**95**		
Pune	Afternoon	LINE-7	Dec 5, 2024	12	94	94	1.56666667
LINE-7 - Total					**94**		
LINE-7 - Total					**94**		
12 - Total					**94**		
12 - Total					**94**		
94 - Total					**94**		
94 - Total					**94**		
Ahmedabad	Afternoon	LINE-7	Jul 24, 2024	7	91	91	1.51666667
LINE-7 - Total					**91**		
LINE-7 - Total					**91**		
7 - Total					**91**		
7 - Total					**91**		
91 - Total					**91**		
91 - Total					**91**		
Pune	Afternoon	LINE-8	Jan 23, 2024	1	85	85	1.41666667
LINE-8 - Total					**85**		
LINE-8 - Total					**85**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
1 - Total					**85**		
1 - Total					**85**		
85 - Total					**85**		
85 - Total					**85**		
Ahmedabad	Afternoon	LINE-8	Nov 17, 2024	11	82	82	1.36666667
LINE-8 - Total					**82**		
LINE-8 - Total					**82**		
11 - Total					**82**		
11 - Total					**82**		
82 - Total					**82**		
82 - Total					**82**		
Pune	Afternoon	LINE-4	Apr 16, 2024	4	74	74	1.23333333
LINE-4 - Total					**74**		
LINE-4 - Total					**74**		
4 - Total					**74**		
4 - Total					**74**		
74 - Total					**74**		
74 - Total					**74**		
Ahmedabad	Afternoon	LINE-4	Dec 17, 2024	12	68	68	1.13333333
LINE-4 - Total					**68**		
LINE-4 - Total					**68**		
12 - Total					**68**		
12 - Total					**68**		
68 - Total					**68**		
68 - Total					**68**		
Ahmedabad	Afternoon	LINE-1	Feb 22, 2024	2	63	63	1.05

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
LINE-1 - Total				2	**63**		
LINE-1 - Total					**63**		
2 - Total					**63**		
2 - Total					**63**		
63 - Total					**63**		
63 - Total					**63**		
Pune	Afternoon	LINE-8	Aug 12, 2024	8	62	62	1.03333333
LINE-8 - Total					**62**		
LINE-8 - Total					**62**		
8 - Total					**62**		
8 - Total					**62**		
62 - Total					**62**		
62 - Total					**62**		
Pune	Afternoon	LINE-2	Aug 27, 2024	8	59	59	0.98333333
LINE-2 - Total					**59**		
LINE-2 - Total					**59**		
8 - Total					**59**		
8 - Total					**59**		
59 - Total					**59**		
59 - Total					**59**		
Ahmedabad	Afternoon	LINE-7	Apr 8, 2024	4	54	54	0.9
LINE-7 - Total					**54**		
LINE-7 - Total					**54**		
4 - Total					**54**		
4 - Total					**54**		
54 - Total					**54**		
54 - Total					**54**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
Pune	Afternoon	LINE-3	Sep 6, 2024	9	42	42	0.7
LINE-3 - Total					**42**		
LINE-3 - Total					**42**		
9 - Total					**42**		
9 - Total					**42**		
42 - Total					**42**		
42 - Total					**42**		
Ahmedabad	Afternoon	LINE-2	May 27, 2024	5	40	40	0.66666667
LINE-2 - Total					**40**		
LINE-2 - Total					**40**		
5 - Total					**40**		
5 - Total					**40**		
40 - Total					**40**		
40 - Total					**40**		
Ahmedabad	Afternoon	LINE-1	Apr 10, 2024	4	33	33	0.55
LINE-1 - Total					**33**		
LINE-1 - Total					**33**		
4 - Total					**33**		
4 - Total					**33**		
33 - Total					**33**		
33 - Total					**33**		
Pune	Afternoon	LINE-6	Dec 24, 2024	12	32	32	0.53333333
LINE-6 - Total					**32**		
LINE-6 - Total					**32**		
12 - Total					**32**		
12 - Total					**32**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
32 - Total					**32**		
32 - Total					**32**		
Ahmedabad	Afternoon	LINE-8	Feb 1, 2024	2	30	30	0.5
Pune	Afternoon		Feb 27, 2024		30	30	0.5
LINE-8 - Total					**60**		
LINE-8 - Total					**60**		
2 - Total					**60**		
2 - Total					**60**		
30 - Total					**60**		
30 - Total					**60**		
Ahmedabad	Afternoon	LINE-6	Feb 29, 2024	2	18	18	0.3
LINE-6 - Total					**18**		
LINE-6 - Total					**18**		
2 - Total					**18**		
2 - Total					**18**		
18 - Total					**18**		
18 - Total					**18**		
Pune	Afternoon	LINE-1	Feb 24, 2024	2	17	17	0.28333333
LINE-1 - Total					**17**		
LINE-1 - Total					**17**		
2 - Total					**17**		
2 - Total					**17**		
17 - Total					**17**		
17 - Total					**17**		
Pune	Afternoon	LINE-4	May 22, 2024	5	15	15	0.25
LINE-4 - Total					**15**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
LINE-4 - Total				5	**15**		
5 - Total					**15**		
5 - Total					**15**		
15 - Total					**15**		
15 - Total					**15**		
Ahmedabad	Afternoon	LINE-3	May 10, 2024	5	12	12	0.2
LINE-3 - Total					**12**		
LINE-3 - Total					**12**		
5 - Total					**12**		
5 - Total					**12**		
12 - Total					**12**		
12 - Total					**12**		
Pune	Afternoon	LINE-1	Mar 6, 2024	3	11	11	0.18333333
LINE-1 - Total					**11**		
LINE-1 - Total					**11**		
3 - Total					**11**		
3 - Total					**11**		
11 - Total					**11**		
11 - Total					**11**		
Ahmedabad	Afternoon	LINE-1	Oct 1, 2024	10	9	9	0.15
LINE-1 - Total					**9**		
LINE-1 - Total					**9**		
10 - Total					**9**		
10 - Total					**9**		
9 - Total					**9**		
9 - Total					**9**		
Overall - Total					**4,953**		

Total Production Hours Lost Due to Downtime:
82.55

Plant Location	Shift	Assembly Line	Production Date	Report Month	Downtime (mins)	Downtime (mins)	Total Production Hours Lost
Overall - Total					**4,953**		

Total Production Hours Lost Due to Downtime:
82.55